

Class 9 Science – Chapter 11: Sound

Conceptual Foundations, Characteristics, Applications & Numericals

All answers below are **exam-ready**, written strictly as per **CBSE marking scheme**, and include **expected keywords** that should be underlined in exams.

1. Explain how a school bell produces sound.

Answer: When a school bell is struck, it starts **vibrating**. These vibrations produce alternate **compressions and rarefactions** in the surrounding air. The disturbance travels through the air as a **sound wave**, which reaches our ears and is perceived as sound.

Expected Keywords: vibration, compressions, rarefactions, sound waves, air

2. Why are sound waves categorized as mechanical waves?

Answer: Sound waves require a **material medium** such as a solid, liquid, or gas to travel. They cannot propagate through a **vacuum**. Since sound travels by the **vibration of particles of the medium**, it is classified as a **mechanical wave**.

Expected Keywords: material medium, vacuum, vibration of particles, mechanical wave

3. Why is a sound wave considered a longitudinal wave?

Answer: In a sound wave, the particles of the medium vibrate **parallel to the direction of propagation** of the wave. This produces alternate regions of **compression (C)** and **rarefaction (R)**. Hence, sound waves are **longitudinal waves**.

Expected Keywords: parallel motion, compressions, rarefactions, longitudinal

4. Describe an experiment illustrating that sound requires a material medium to travel.

Answer: In the **bell jar experiment**, an electric bell is placed inside a glass jar connected to a vacuum pump. Initially, sound is heard clearly. As air is pumped out, the sound becomes faint and finally inaudible,

though the bell continues to vibrate. This proves that **sound cannot travel in a vacuum** and requires a **material medium**.

Expected Keywords: bell jar experiment, vacuum pump, no sound in vacuum, medium

5. Would you be able to hear sound on the moon? Why or why not?

Answer: No, sound cannot be heard on the moon because there is **no atmosphere**. Since sound is a **mechanical wave** and requires a **material medium** to propagate, it cannot travel on the moon.

Expected Keywords: no atmosphere, no medium, mechanical wave

6. Compare the speed of sound in air, water, and iron.

Answer: The speed of sound is **slowest in gases (air)**, faster in **liquids (water)**, and **fastest in solids (iron)**. This is because solids have **greater elasticity** and closely packed particles.

Expected Keywords: solids > liquids > gases, elasticity, particle spacing

7. Explain why lightning is seen before thunder is heard.

Answer: Light travels at a speed of $3 \times 10^8 \text{ m/s}$, whereas sound travels at about **344 m/s** in air. Since light travels much faster than sound, lightning is seen before thunder is heard.

Expected Keywords: speed of light, speed of sound, faster propagation

8. What minimum distance is needed to hear a distinct echo?

Answer: The persistence of hearing is **0.1 s**. In this time, sound travels: Distance = $344 \times 0.1 = 34.4 \text{ m}$ (to and fro) Hence, minimum distance from the reflecting surface = $34.4 / 2 = 17.2 \text{ m}$.

Expected Keywords: echo, persistence of hearing, 0.1 s, 17.2 m

9. Differentiate between loudness and intensity of sound.

Answer: - **Loudness** is a **subjective sensation** that depends on the **sensitivity of the human ear**. - **Intensity** is an **objective physical quantity** defined as sound power per unit area and is measured in **W/m²**.

Expected Keywords: subjective, objective, ear sensitivity, W/m^2

10. Define reverberation and suggest ways to reduce it in large halls.

Answer: Reverberation is the **persistence of sound** caused by **multiple reflections** from walls and ceilings. It can be reduced by using **sound-absorbing materials** such as heavy curtains, carpets, cushioned seats, and compressed fibreboards.

Expected Keywords: multiple reflections, persistence of sound, sound absorption

11. Define wavelength, frequency, time period, and amplitude.

Answer: - **Wavelength (λ):** Distance between two consecutive compressions or rarefactions (unit: **metre**). - **Frequency (f):** Number of vibrations per second (unit: **hertz**). - **Time Period (T):** Time taken to complete one vibration (unit: **second**). - **Amplitude:** Maximum displacement of particles from mean position (unit: **metre**).

Expected Keywords: units, vibrations, displacement, mean position

12. Identify the wave properties that determine loudness and pitch.

Answer: - Loudness depends on **Amplitude**. - Pitch depends on **Frequency**.

Expected Keywords: amplitude, frequency

13. Which sound characteristic helps in identifying a friend's voice in the dark?

Answer: The **quality or timbre** of sound helps us distinguish between sounds of the same pitch and loudness.

Expected Keywords: quality, timbre, waveform

14. What is the typical range of frequencies that humans can hear?

Answer: The audible range of human hearing is **20 Hz to 20,000 Hz (20 kHz)**.

Expected Keywords: audible range, 20 Hz, 20 kHz

15. Define infrasonic and ultrasonic sounds with examples.

Answer: - **Infrasonic sound:** Frequency less than **20 Hz** (e.g., elephants, earthquakes). - **Ultrasonic sound:** Frequency greater than **20,000 Hz** (e.g., bats, dolphins).

Expected Keywords: below 20 Hz, above 20 kHz, examples

16. Describe how SONAR works and its uses.

Answer: SONAR works on the principle of **echolocation**. Ultrasonic waves are transmitted into water, which reflect from the seabed or underwater objects and return as echoes. The **time interval** is used to calculate depth or distance.

Expected Keywords: echolocation, ultrasonic waves, reflection, echo

17. How is ultrasound used for cleaning and detecting flaws?

Answer: Ultrasound is used for cleaning by producing **high-frequency vibrations** in a liquid, which remove dirt from small crevices. For flaw detection, ultrasonic waves reflect back from **cracks or cavities** inside metal blocks.

Expected Keywords: high-frequency vibrations, cracks, reflection

18. Explain the function of the human ear.

Answer: The human ear has three parts: - **Outer ear:** Collects sound waves. - **Middle ear:** Amplifies vibrations using three bones (hammer, anvil, stirrup). - **Inner ear:** Cochlea converts vibrations into **electrical signals** sent to the brain.

Expected Keywords: outer ear, middle ear, cochlea, electrical signals

19. A sound source produces 100 compressions and 100 rarefactions in 2 seconds. Find the frequency.

Answer: One compression and one rarefaction form **one vibration**. Number of vibrations = 100
Frequency = Vibrations / Time = $100 / 2 = 50 \text{ Hz}$

Expected Keywords: vibration, frequency formula, Hz

20. Determine the distance to a cliff if a gun's echo is heard after 2 seconds.

(Given: Speed = 340 m/s) **Answer:** $2d = v \times t$ $2d = 340 \times 2$ $d = \mathbf{340\ m}$

Expected Keywords: echo, reflection, divide by 2, distance

 **Final Examiner Tip:**

Underline **keywords**, write **units**, and follow **stepwise presentation** for full marks.